Master of Business Analytics

Prescriptive Analytics & Optimization

MBAN 5540

Class 03
Linear Programming Sensitivity Analysis
Python + Gurobi

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Review

- What we know so far:
- How to use mathematical approach to model business problems using linear programming.
- How to solve LP using graphical solution (limited to 2 decision variables).
- How to use Excel Solver to model and solve LP problems.

Flair Company Example

- Recall the Flair company example:

$$
\begin{aligned}
Maximize\ \ 7T + 5C \\
3T + 4C &\leq 2{,}400 \\
2T + 1C &\leq 1{,}000 \\
C &\leq 450 \\
T &\geq 100 \\
T &\geq 0 \\
C &\geq 0
\end{aligned}
$$

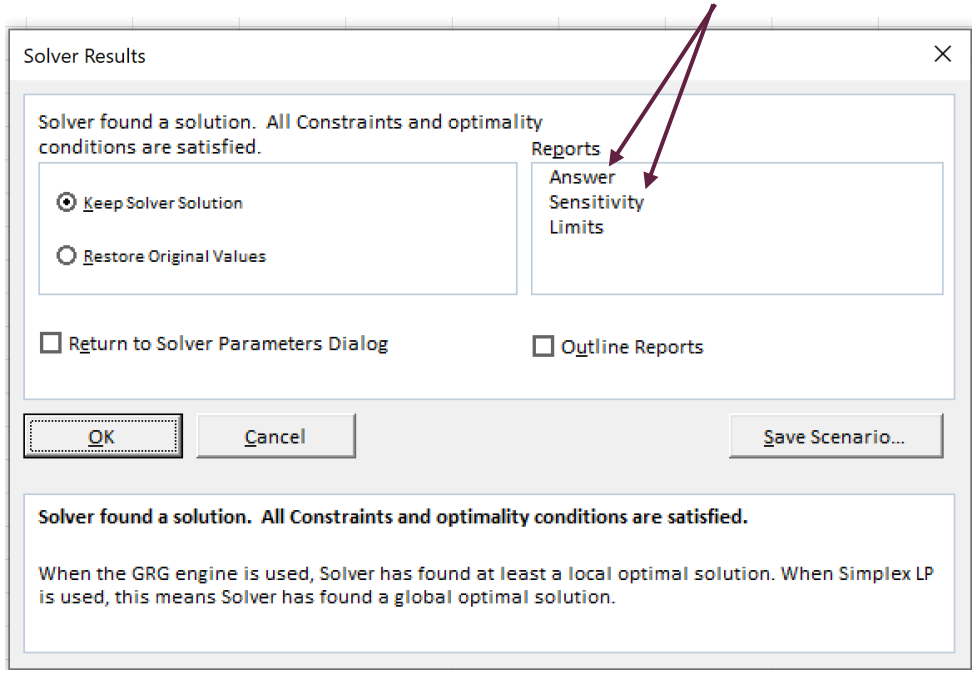
Solver Reports

- When Excel Solver solves the mathematical model, it generates two reports:

- Answer Report

- Sensitivity Report

Click on these two;
then click OK.



Answer Report

Objective Cell (Max)

Cell	Name	Original Value	Final Value
$D$6	Objective	0	4040

Variable Cells

Cell	Name	Original Value	Final Value	Integer
$B$5	Decision Variables: T	0	320	Contin
$C$5	Decision Variables: C	0	360	Contin

Constraints

Cell	Name	Cell Value	Formula	Status	Slack
$D$11	Min Table LHS	320	$D$11>=$F$11	Not Binding	220
$D$8	Carpentry LHS	2400	$D$8<=$F$8	Binding	0
$D$9	Painting LHS	1000	$D$9<=$F$9	Binding	0
$D$10	Max Chair LHS	360	$D$10<=$F$10	Not Binding	90

Answer Report: Objective Cell

Objective Cell (Max)

Cell	Name	Original Value	Final Value
$D$6	Objective	0	4040

- The Objective Cell gives us information obtained from the objective function.
  - Cell: which cell the objective function is in.
  - Original Value: the initial value of objective function before solver was initiated
  - Final Value: the optimal value of objective function which is obtained by replacing the optimal values of the decision variables (optimal corner point) in the objective function.

Answer Report: Variable Cells

Variable Cells

Cell	Name	Original Value	Final Value	Integer
$B$5	Decision Variables: T	0	320	Contin
$C$5	Decision Variables: C	0	360	Contin

- The Variable Cells table gives us information obtained from the decision variables.
  - Cell: which cells the decision variables are located in.
  - Original Value: the initial value of decision variables before solver was initiated.
  - Final Value: the optimal value of decision variables (optimal corner point).

Answer Report: Constraints

Constraints

Cell	Name	Cell Value	Formula	Status	Slack
$D$11	Min Table LHS	320	$D$11>=$F$11	Not Binding	220
$D$8	Carpentry LHS	2400	$D$8<=$F$8	Binding	0
$D$9	Painting LHS	1000	$D$9<=$F$9	Binding	0
$D$10	Max Chair LHS	360	$D$10<=$F$10	Not Binding	90

- The Constraints table gives us information obtained from the constraints.
  - Cell: which cells the LHS (Left hand side) of the constraints are in.
  - Cell Value: the final value of the LHS of the constraint at the optimal solution (how much of the given resource is used in the optimal solution).
  - Formula: the formula specified in Solver for the constraints.

Answer Report 3: Constraints

Constraints

Cell	Name	Cell Value	Formula	Status	Slack
$D$11	Min Table LHS	320	$D$11>=$F$11	Not Binding	220
$D$8	Carpentry LHS	2400	$D$8<=$F$8	Binding	0
$D$9	Painting LHS	1000	$D$9<=$F$9	Binding	0
$D$10	Max Chair LHS	360	$D$10<=$F$10	Not Binding	90

- Status: indicates whether the constraint is **binding** or **Not binding**.
  - Binding means the constrain is exactly satisfied at the optimal solution (i.e., LHS = RHS).
  - Not Binding means LHS ≠ RHS.
  - When the optimal corner point is on a constraint, that constraint is binding!

Answer Report 3: Constraints

Constraints

Cell	Name	Cell Value	Formula	Status	Slack
$D$11	Min Table LHS	320	$D$11>=$F$11	Not Binding	220
$D$8	Carpentry LHS	2400	$D$8<=$F$8	Binding	0
$D$9	Painting LHS	1000	$D$9<=$F$9	Binding	0
$D$10	Max Chair LHS	360	$D$10<=$F$10	Not Binding	90

= 320-100

= 450-360

- Slack: Magnitude (absolute value) of the difference between RHS and LHS values of the constraint.
  - For a "≤" constraint, RHS - LHS is called **Slack** (amount of resources that is left unused).
  - For a "≥" constraint, LHS - RHS is called **Surplus** (the extent to which the constraint is over satisfied).
  - If a constraint is binding, Slack = Surplus = 0.
  - If a constraint is not binding, Slack > 0 or Surplus > 0.

Why Sensitivity Analysis?

- Certainty assumption for LP is not true for most real-world problems.
- Conditions are dynamic and changing.

- Two main reason:
  - Error in the estimation of input data/parameters (coefficients and RHS)
  - Managers are interested in What-if analysis

- Shows how sensitive the solution is to change in each input data
- Determines a range for each input data in which the solution is optimal.

- We examine:
  - Changes in objective function coefficients (OFC)
  - Changes in RHS

Sensitivity Report

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
$B$5	Decision Variables: T	320	0	7	3	3.25
$C$5	Decision Variables: C	360	0	5	4.333333333	1.5

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
$D$11	Min Table LHS	320	0	100	220	1E+30
$D$8	Carpentry LHS	2400	0.6	2400	225	900
$D$9	Painting LHS	1000	2.6	1000	600	150
$D$10	Max Chair LHS	360	0	450	1E+30	90

Sensitivity Report: Variable Cells

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
$B$5	Decision Variables: T	320	0	7	3	3.25
$C$5	Decision Variables: C	360	0	5	4.333333333	1.5

- Final Value: optimal values of decision variable.
- Objective Coefficient: Coefficients of objective function, which can be used to find the objective function value $(7(320) + 5(360) = 4{,}040)$.
- Range of Optimality is the interval for each coefficient in the objective function over which, the optimal values of all decision variables remain unchanged.
- Allowable Increase/Decrease help us to find the range of optimality.
- Allowable Increase/Decrease are the range of values for which we can change the *OFC*s, and current Corner Point will remain as Optimal Solution.

Changes to Objective Function Coefficients

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
$B$5	Decision Variables: T	320	0	7	3	3.25
$C$5	Decision Variables: C	360	0	5	4.333333333	1.5

- Range of Optimality:
  - For T with coefficient 7: $(7 - 3.25\,, 7 + 3) = (3.75\,, 10)$
  - For C with coefficient 5: $(5 - 1.5\,, 5 + 4.33) = (3.5, 9.33)$

  - As long as T and C are in the above range, the optimal solution is $(T, C) = (320,360)$

- Check different values for T and C to see the result (one at a time!).

- At what profit do we produce less chairs?

Sensitivity Report: Constraints

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
$D$11	Min Table LHS	320	0	100	220	1E+30
$D$8	Carpentry LHS	2400	0.6	2400	225	900
$D$9	Painting LHS	1000	2.6	1000	600	150
$D$10	Max Chair LHS	360	0	450	1E+30	90

- The Constraints table presents information on impact of changes in *RHS.*
  - Final Value: the values of how much of each resource (constraint) is used in the optimal solution (the value of LHS in optimal solution).
  - Constraint RHS Side: the value input for the RHS of each constraint equation.
  - The above can be used to determine if a constraint is binding or not.

Sensitivity Report: Constraints

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
$D$11	Min Table LHS	320	0	100	220	1E+30
$D$8	Carpentry LHS	2400	0.6	2400	225	900
$D$9	Painting LHS	1000	2.6	1000	600	150
$D$10	Max Chair LHS	360	0	450	1E+30	90

- Shadow Price: is the change in optimal objective function value for **one unit** of change in RHS.
- The shadow price for a non-binding constraint is $= 0$
- The shadow price for a binding constraint is $\neq 0$

Change in the RHS

- For a "≤" constraint:
  - Increasing the RHS → Larger feasible region
  - Decreasing the RHS → Smaller feasible region

- For a "≥" constraint:
  - Increasing the RHS → Smaller feasible region
  - Decreasing the RHS → Larger feasible region

- When we make the feasible region larger, the solution cannot be worse.

- When we make the feasible region smaller, the solution cannot be better.

Sensitivity Report: Constraints

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
$D$11	Min Table LHS	320	0	100	220	1E+30
$D$8	Carpentry LHS	2400	0.6	2400	225	900
$D$9	Painting LHS	1000	2.6	1000	600	150
$D$10	Max Chair LHS	360	0	450	1E+30	90

- If we change 2,400 to 2,401
  - we make the feasible region larger
  - the objective function value will be $4{,}040 + 0.6 = 4{,}040.6$

- If we change 2,400 to 2,399
  - we make the feasible region smaller
  - the objective function value will be $4{,}040 - 0.6 = 4{,}039.4$

Sensitivity Report: Constraints

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
$D$11	Min Table LHS	320	0	100	220	1E+30
$D$8	Carpentry LHS	2400	0.6	2400	225	900
$D$9	Painting LHS	1000	2.6	1000	600	150
$D$10	Max Chair LHS	360	0	450	1E+30	90

- If we change 1,000 to 1,010
  - we make the feasible region larger
  - the objective function value will be $4{,}040 + 10(2.6) = 4{,}066$

- If we change 1,000 to 990
  - we make the feasible region smaller
  - the objective function value will be $4{,}040 - 10(2.6) = 4{,}014$

- What if we change 100 and 450?

Sensitivity Report: Constraints

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
$D$11	Min Table LHS	320	0	100	220	1E+30
$D$8	Carpentry LHS	2400	0.6	2400	225	900
$D$9	Painting LHS	1000	2.6	1000	600	150
$D$10	Max Chair LHS	360	0	450	1E+30	90

- Validity of Shadow Price for a Constraint:
  - The **range of feasibility** (for a constraint) is the interval over which the RHS value of a constraint can vary while its shadow price remains constant.
  - This can be determined by Allowable Decrease and Allowable Increase interval.
  - For nonbinding constraints, the shadow price is zero and one of Allowable Decrease or Allowable Increase is **infinity**, shown as 1E+30 in Solver.

Use of Shadow Price

1. Shows us the value of additional resources:
   - The value of one extra hour in carpentry Dept. is \$0.6
   - The value of one extra hour in painting Dept. is \$2.6

2. New product Evaluation:
   - Shows the opportunity cost of producing a new product that uses the same resources
   - Example: A new product (e.g., cabinet) that needs 3 hours/unit in the carpentry department and 2 hours/unit in the painting department, and a profit margin of \$8 /unit.

   - Cost of producing (marginal value):
     - $3(0.6) + 2(2.6) + 0(0) + 0(0) = \$7 \quad < \quad \$8$

   - We should produce it !

Use of Shadow Price

3. Identify how to spend additional resource:
   - If we have additional money that can be used to increase the number hours in carpentry or painting department, how do we allocate the money?
   - The constraint with higher shadow price will be our priority.
     - Every hour in carpentry department will increase our profit by $0.6.
     - Every hour in painting department will increase our profit by $2.6.
   - We should pay attention to allowable range.

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
$D$11	Min Table LHS	320	0	100	220	1E+30
$D$8	Carpentry LHS	2400	0.6	2400	225	900
$D$9	Painting LHS	1000	2.6	1000	600	150
$D$10	Max Chair LHS	360	0	450	1E+30	90

Reduced Cost

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
$B$5	Decision Variables: T	320	0	7	3	3.25
$C$5	Decision Variables: C	360	0	5	4.333333333	1.5

- Reduced Cost: is the shadow price of the non-negativity constraints.
- A non-negativity constraint: $x \geq 0$
  - If a non-negativity constraint is binding (e.g., $x = 0$), then the reduced cost is > 0.
  - If a non-negativity constraint is nonbinding (e.g., $x > 0$), then the reduced cost is 0.
  - Therefore, we always have: $x.(\text{reduce cost}) = 0$

- When reduced cost is <u>not zero</u>, it means that we <u>do not produce</u> that variable.

Reduce Cost

- In our example, let's change OFC of tables to 20 (from 7) and re-solve the model:

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
$B$5	Decision Variables: T	500	0	20	1E+30	10
$C$5	Decision Variables: C	0	-5	5	5	1E+30

- The production of C will change to 0.
- Reduced cost of C will change to -5.

- How much improvement do we need for the profit of chairs so that we produce it again? (we call this entering the solution).

Reduced Cost – Interpretation 1

- The reduced cost (for a decision variable in a non-negativity constraint) is the amount of improvement for the objective function coefficient of that decision variable needed for the decision variable to "enter" the solution and get some positive value.

  - Reduced cost of C is -5.
  - Improvement to make C to enter the solution:
    - $5 - (-5) = 10$

  - Check the solution with a new value > 10 for C coefficient

Reduced Cost – Interpretation 2

- When the coefficient of C is 5, it is not profitable to produce any chairs.
- In previous interpretation, we saw that the profit of C needs to be at least 10 so that it is profitable to produce chairs.
- Now, what if we force the model to produce chairs?
- Reduced cost of C is -5. For every chair that we produce, we lose \$5 of our profit!
- Reduced Cost: is the shadow price of the non-negativity constraints.

Change in RHS Flowchart

Change in RHS by x unit

↓

Change within the allowable range?

NO → Solver report is not valid anymore. Solve the model again and generate the new reports.

YES ↓

Is the constraint binding?

NO → No change to the solution

YES ↓

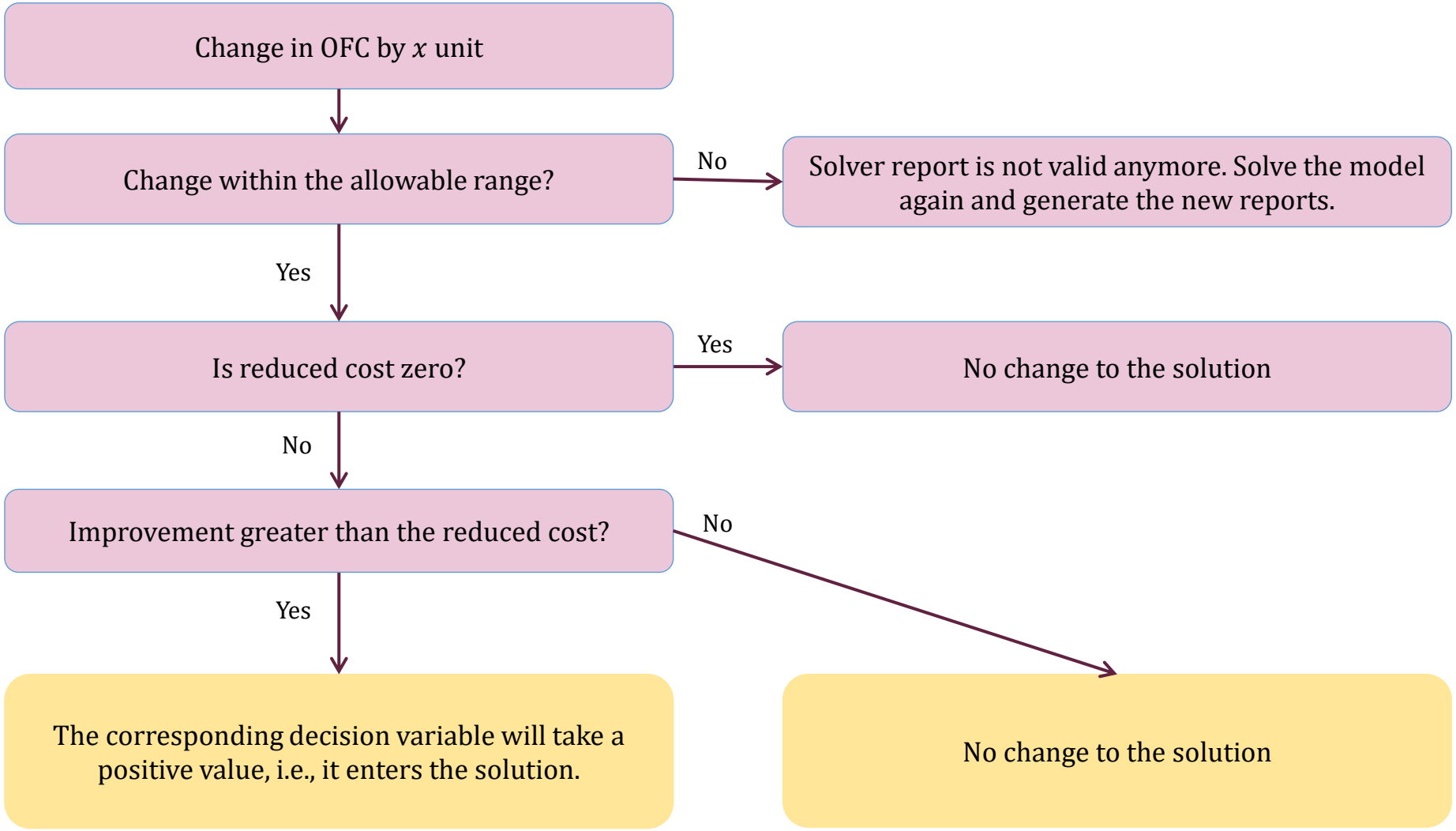
Does this make the feasible region larger or smaller?

Larger ↓ Objective function value will be **improved** by: x (Shadow price of the constraint)

Smaller → Objective function value will be **disimproved** by: x (Shadow price of the constraint)

	Min	Max
Improve	-	+
Disimprove	+	-

Change in OFC Flowchart



A Minimization Example

- Diet Drink Company

	Ingredients				
	A	B	C	D	Requirement
Chemical X	3	4	8	10	At least 280 units
Chemical Y	5	3	6	6	At least 200 units
Chemical Z	10	25	20	40	At most 1,050 units
Cost/ounce	$40	$20	$60	$30	

- Min daily ingredient dosage of 36 and Max of 40.

A Minimization Example

- Diet Drink Company

$$Min\ \ 40A + 20B + 60C + 30D$$

$$\begin{aligned}
A + B + C + D &\geq 36 \\
3A + 4B + 8C + 10D &\geq 280 \\
5A + 3B + 6C + 6D &\geq 200 \\
10A + 25B + 20C + 40D &\leq 1{,}050 \\
A + B + C + D &\leq 40 \\
A, B, C, D &\geq 0
\end{aligned}$$